

CAPE TOWN · ENERGY · DATA · AI

Tiffany Woodley

Lead Data Scientist

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Professional Profile

Engineer and data scientist who builds things people actually use — from renewable energy modelling to enterprise-scale AI, ready to bring that craft back to the energy sector.

I'm an engineer and data scientist, and what I love most is building things people actually use. I start with the decision someone needs to make, then work backwards: shaping the solution, leading the team and owning the result.

I began in renewable energy, building SolaFuture's first battery sizing tool, which became its standard proposal engine. At Solink, I led the battery modelling approach through the load-shedding uncertainty, then got promoted to head up systems development, taking monitoring, automation and consulting platforms through all the stages.

Now, at Sand Technologies, I lead data science and engineering on multiple projects, building the pipelines, MLOps and monitoring that keep AI systems dependable. I moved into enterprise on purpose, to learn how to build at scale, and I'm ready to bring that back to the energy sector.

Energy and modelling

Battery and Solar Financial Modeling
Battery System Control Systems
Forecasting
Optimisation

Data science and AI

Machine learning
Generative AI
MLOps

Technology

Software engineering
Data engineering
Solutions Architecture
Technical Leadership

Timeline

2025 – Now
Full time

Senior Data Scientist · Sand Technologies

Healthcare – Lead Product Data Science and Data Engineering

- Led a team of senior data engineers to build reusable data pipelines deployed across multiple healthcare ministry analytical systems.
- Led a project to build a structured way of building structured data models that give analytics and AI a reliable foundation to reason on.
- Lead data science and data engineering projects: framing the problem, choosing the approach and delivering with cross-functional teams.
- Translated messy operational requirements into practical, production-ready data solutions.

Central – Data Science and AI

- Focus on the framework around forecasting — how to take different modelling approaches into production systems.
- Worked on components for reusable ML pipelines so teams across different projects and industries can build on shared foundations rather than starting from scratch.
- Built bespoke tools and applications for data quality and data drift monitoring, designed to sit directly inside pipelines and workflows so issues are caught early before they affect model performance in production.
- Ran experiments on AI-assisted coding to get speed without spaghetti code — using strong software principles so pieces slot straight into a pipeline — and built proofs of concept for client demos across industries.
- Helped shape how generative AI is brought into people's workflows so they can build scalable solutions, with worked examples for industry workflows.

Software engineering | Machine learning | Data engineering | Forecasting | Generative AI | MLOps | Solutions Architecture | Data quality | Model monitoring | Technical Leadership

2024 – 2025
Full time

Head of Systems Development · Solink

- Led the software team, setting technical direction and standards across all products and internal platforms.
- Introduced processes for product and technical delivery, so projects were technically planned based on client needs.
- Owned every technical solution, from the live products to the roadmap for new ones.
- Built internal platforms that automated operations, including a reporting framework and energy financial models embedded in day-to-day processes.

Software engineering | Battery and Solar Financial Modeling | Solutions Architecture | Technical Leadership | Node.js | Postgres | Appscript | AWS

2023 – 2024
Full time

Systems Engineer · Solink

- Built the modelling strategy the company adopted when there was no established way to model battery projects under load-shedding uncertainty — using EskomSePush schedules, seasonality and multi-year scenario paths to show each project's true risk if load-shedding eased or disappeared.
- When shedding eased, the same framework switched to other dispatch strategies such as arbitrage and grid avoidance, so clients saw full financial sensitivity, not just short-term returns.
- Guided sizing with the models, showing that oversizing for a handful of extreme events quickly hit diminishing returns, and where the practical cut-off sat.
- Led this approach across the team, mentored engineers in it and reviewed every proposal before it went out.
- Optimised paid-for solar forecast operating points to reduce costs and ran system-size optimisation.
- Applied dispatch knowledge in practice — writing the control algorithms behind the energy models and connecting dispatch logic to the financial and forecasting systems around them.
- Built solar production forecasting designed to plug into the monitoring software and flag underperformance, and left it ready for integration on the company's development backlog.
- Rearchitected the automated energy reporting platform to improve the efficiency so that analysts had enough time to get reports out during the reporting cycle.
- Tightened delivery cycles by introducing agile ways of working across the engineering team.

Software engineering | Battery and Solar Financial Modeling | Battery System Control Systems | Forecasting | Machine learning | Optimisation | Technical Leadership | Solutions Architecture | Node.js | Postgres | Appscript

2022 – 2023
Full time

Machine Learning Engineer · Bytefuse.ai

- Built reinforcement-learning controllers for traffic signals to improve flow and reduce congestion.
- Developed traffic prediction models.

Software engineering | Machine learning | Forecasting | Optimisation | Reinforcement learning

2020 – 2022
Education

MSc Advanced Analytics · University of Cape Town

- Thesis: reinforcement learning for peak shaving — treating battery control as a real optimisation problem with costs, constraints and delayed rewards.
- After finishing the Data Science coursework, I moved on to Advanced Analytics for more statistical depth.
- Advanced statistics and modelling: multivariate statistics, regression, simulation, optimisation and machine learning.

Machine learning | Optimisation | Battery System Control Systems | Statistics | Reinforcement learning

2019 – 2020
Education

MSc Data Science · University of Cape Town

- Strong statistical foundation in supervised and unsupervised learning, industry data science, databases and visualisation.

Machine learning | Data engineering | Data visualisation | Databases

2017 – 2019
First roles

Junior Software Engineer · SolaFuture Energy

Intern in 2017, then Junior Software Engineer 2018–2019.

- Started as an intern while finishing my degree; wrote software that simulated battery operation hour by hour and realised the same engine could size battery and solar systems optimally.
- Built a Python backend linked to the company's Excel financial models that looped over system configurations and produced design and sizing reports.
- Grew that work into SolaFuture's first battery sizing software — a Python-backed financial model that simulated projects, ran scenarios and generated every client battery proposal.
- The tool became the company's standard proposal engine and stayed in use after I left.
- Built a peak-shaving model — my first experience of battery control.
- Built the load forecasting that drove the peak-shaving algorithms, predicting demand so battery control could act at the right moments.

Software engineering | Battery and Solar Financial Modeling | Battery System Control Systems | Forecasting | VBA | Linux server | SQL

2012 – 2018
Education

BEng Electrical & Electronic Engineering · Stellenbosch University

Electrical and electronic engineering foundation — the base for later work in energy systems and data science.

Engineering | Systems

Open to conversations about energy systems, applied AI and building things that hold up in production. · tiff.woodley@gmail.com